Bureau
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# Le système international d'unités
# The International System of Units

9 édition 2003

Annexe 2
Annex 2

## Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

## Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

## Iodine ()

## Iodine ()




CCL-CCTF Frequency Standards Working Group

**Bureau International**
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# Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
# Part 2: Iodine ()

**CCL-CCTF Frequency Standards Working Group**

**9th edition 2003**

Appendix 2

# 1. CIPM recommended values

The values $f = 473\ 612\ 353\ 604\ \text{kHz}$
$\lambda = 632\ 991\ 212.58\ \text{fm}$

with a relative standard uncertainty of $2.1 \times 10^{-11}$ apply to the radiation of a He-Ne laser with an internal iodine cell, stabilized using the third harmonic detection technique, subject to the conditions:

- cell-wall temperature $(25 \pm 5)\ °C$[(1)];
- cold-finger temperature $(15.0 \pm 0.2)\ °C$;
- frequency modulation width, peak-to-peak, $(6.0 \pm 0.3)$ MHz;
- one-way intracavity beam power (i.e. the output power divided by the transmittance of the output mirror) $(10 \pm 5)$ mW for an absolute value of the power shift coefficient $\leq 1.0\ \text{kHz mW}^{-1}$.

[(1)] For the specification of operating conditions, such as temperature, modulation width and laser power, the symbols ± refer to a tolerance, not an uncertainty.

These conditions are by themselves insufficient to ensure that the stated standard uncertainty will be achieved. It is also necessary for the optical and electronic control systems to be operating with the appropriate technical performance. The iodine cell may also be operated under relaxed conditions, leading to the larger uncertainty specified in section 2 below.

# 2. Source data

Adopted value

$f = 473\ 612\ 353\ 604\ (10)$ kHz
$u_c/y = 2.1 \times 10^{-11}$

for which:
$\lambda = 632\ 991\ 212.579\ (13)$ fm
$u_c/y = 2.1 \times 10^{-11}$

calculated from

| $f$/ kHz | $u_c/y$ | source data |
|---|---|---|
| 8.2 | $4.0 \times 10^{-12}$ | [1], [2] |
| 7.4 | $3.0 \times 10^{-12}$ | [1], [3] |
| 4.2 | $1.4 \times 10^{-11}$ | See Section 2.1 |
| 8.2 | $5.3 \times 10^{-12}$ | [5] |
| Unweighted mean: | $(f_{\mathrm{BIPM4}} - f_{\mathrm{CIPM97}}) = 7.0$ kHz | |

The source data are all given with respect to the BIPM4 laser standard frequency. The relative standard uncertainty includes the uncertainty in the absolute frequency measurement and the uncertainty obtained by comparing the different frequency standards with the BIPM4 standard. The CCL proposed that the recommended radiation for the R(127) 11-5 transition, using 633 nm He-Ne lasers, no longer correspond to the $a_{13}$ or i component, but is replaced by the $a_{16}$ or f component, which was decided by the CIPM 2001.

The CCL adopted a correction of the previous recommended frequency by $+7$ kHz, giving the frequency of the f component to be 473 612 353 604 kHz. The CCL also revised the coefficient of the tolerated one-way intracavity beam power influencing the average uncertainty of beat-frequency measurements between two stabilized lasers. This results in a combined uncertainty of $u_c = 10$ kHz, corresponding to a relative uncertainty of $u_c/y = 2.1 \times 10^{-11}$, see Section 2.2. The grouped laser comparisons from national laboratories undertaken by the BIPM (1993-2000) confirm that the choice of a relative standard uncertainty of $2.1 \times 10^{-11}$ is valid [6], [7], [8], [9], [10], [11], [12], [13], [14]. This series of comparisons is a key comparison BIPM.L-K10 and is reported on the BIPM website http://www.bipm.org/kcdb.

For applications where relaxed tolerances, and the resultant wider uncertainty range are acceptable, a laser operated under the conditions recommended in 1983 [15], [16] would lead to a standard uncertainty of about 50 kHz (or a relative standard uncertainty of $1 \times 10^{-10}$).

Source data

## 2.1.

Sugiyama et al. [14] give

$f_f = 473\ 612\ 353\ 604.3$ kHz
$u_c = 1.7$ kHz
as the frequency of the NRLM-P1 laser standard.

This value indicates that $f_f = f_{\mathrm{CIPM97}} + f_{corr}$ where $f_{corr} = 7.3\ \mathrm{kHz}$.

In a comparison with the BIPM4 laser standard [34], they obtained

$f_f - f_{\mathrm{BIPM4}} = 3.1\ \mathrm{kHz}$
$u_c = 6.4\ \mathrm{kHz}$.

Assuming that this frequency has been maintained since, one obtains

$(f_{\mathrm{BIPM4}} - f_{\mathrm{CIPM97}}) = 4.2\ \mathrm{kHz},\ u_c = 6.6\ \mathrm{kHz}$.

## 2.2.

The uncertainties resulting from variations in operational parameters are listed below.

| Parameter | Recommended value | Tolerance | Coefficient | $u$/ kHz |
|---|---|---|---|---|
| Iodine cell | | | | |
| cell-wall temperature | 25 °C | 5 °C | 0.5 kHz °C$^{-1}$ | 2.5 |
| cold-finger temperature | 15 °C | 0.2 °C | −15 kHz °C$^{-1}$ | 3.0 |
| iodine purity | | | | 5.0 |
| Frequency modulation width peak-to-peak | 6 MHz | 0.3 MHz | −10 kHz MHz$^{-1}$ | 3.0 |
| One-way intracavity beam power | 10 mW | 5 mW | ≤ 1.0 kHz mW$^{-1}$ | 5.0 |
| Beat-frequency measurements between two lasers | | | | 5.0 |
| | | | Combined standard uncertainty $u_c = 10.0$ kHz | |

# 3. Absolute frequency of the other transitions related to those adopted as recommended and frequency intervals between transitions and hyperfine components

These tables replace those published in BIPM Com. Cons. Long., 2001, **10**, 168-173 and *Metrologia*, 2003, **40**, 121-123.

The notation for the transitions and the components is that used in the source references. The values adopted for the frequency intervals are the weighted means of the values given in the references.

For the uncertainties, account has been taken of:

- the uncertainties given by the authors;
- the spread in the different determinations of a single component;
- the effect of any perturbing components;
- the difference between the calculated and the measured values.

In the tables, uc represents the estimated combined standard uncertainty ($1\sigma$).

All transitions in molecular iodine refer to the B-X system.

**Table 1**

$\lambda \approx 633$ nm $^{127}I_2$ R(127) 11-5

| $a_n$ | $x$ | $f(a_n) - f(a_{16})$/ MHz | $u_c$/ MHz | $a_n$ | $x$ | $f(a_n) - f(a_{16})$/ MHz | $u_c$/ MHz |
|---|---|---|---|---|---|---|---|
| $a_2$ | t | -721.8 | 0.5 | $a_{12}$ | j | -160.457 | 0.005 |
| $a_3$ | s | -697.8 | 0.5 | $a_{13}$ | i | -138.892 | 0.005 |
| $a_4$ | r | -459.62 | 0.01 | $a_{14}$ | h | -116.953 | 0.005 |
| $a_5$ | q | -431.58 | 0.05 | $a_{15}$ | g | -13.198 | 0.005 |
| $a_6$ | p | -429.18 | 0.05 | $a_{16}$ | f | 0 | — |
| $a_7$ | o | -402.09 | 0.01 | $a_{17}$ | e | 13.363 | 0.005 |
| $a_8$ | n | -301.706 | 0.005 | $a_{18}$ | d | 26.224 | 0.005 |
| $a_9$ | m | -292.693 | 0.005 | $a_{19}$ | c | 144.114 | 0.005 |
| $a_{10}$ | l | -276.886 | 0.005 | $a_{20}$ | b | 152.208 | 0.005 |
| $a_{11}$ | k | -268.842 | 0.005 | $a_{21}$ | a | 161.039 | 0.005 |

Frequency referenced to $a_{16}$ (f), R(127) 11-5, $^{127}I_2$: $f = 473\,612\,353\,604$ kHz [17]

Ref. [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29]

**Table 2**

$\lambda \approx 633$ nm $^{127}I_2$ P(33) 6-3

| $b_n$ | $x$ | $f(b_n)-f(b_{21})$/ MHz | $u_c$/ MHz | $b_n$ | $x$ | $f(b_n)-f(b_{21})$/ MHz | $u_c$/ MHz |
|---|---|---|---|---|---|---|---|
| $b_1$ | u | -922.571 | 0.008 | $b_{12}$ | j | -347.354 | 0.007 |
| $b_2$ | t | -895.064 | 0.008 | $b_{13}$ | i | -310.30 | 0.01 |
| $b_3$ | s | -869.67 | 0.01 | $b_{14}$ | h | -263.588 | 0.009 |
| $b_4$ | r | -660.50 | 0.02 | $b_{15}$ | g | -214.53 | 0.02 |
| $b_5$ | q | -610.697 | 0.008 | $b_{16}$ f | | -179.312 | 0.005 |
| $b_6$ | p | -593.996 | 0.008 | $b_{17}$ | e | -153.942 | 0.005 |
| $b_7$ | o | -547.40 | 0.02 | $b_{18}$ | d | -118.228 | 0.007 |
| $b_8$ | n | -487.074 | 0.009 | $b_{19}$ | c | -36.73 | 0.01 |
| $b_9$ | m | -461.30 | 0.03 | $b_{20}$ | b | -21.980 | 0.007 |
| $b_{10}$ | l | -453.21 | 0.03 | $b_{21}$ | a | 0 | — |
| $b_{11}$ | k | -439.01 | 0.01 | | | | |

Frequency referenced to $a_{16}$ (f), R(127) 11-5, $^{127}I_2$: $f = 473\,612\,353\,604$ kHz [17]

$f(b_{21}, \text{P(33) 6-3}) - f(a_{16}, \text{R(127) 11-5}) = -532.42\,(2)$ MHz [30]

Ref. [25], [30], [31], [32], [33], [34]

**Table 3**

$\lambda \approx 633$ nm $^{129}I_2$ P(54) 8-4

| $a_n$ | $x$ | $f(a_n)-f(a_{28})$/ MHz | $u_c$/ MHz | $a_n$ | $x$ | $f(a_n)-f(a_{28})$/ MHz | $u_c$/ MHz |
|---|---|---|---|---|---|---|---|
| $a_2$ | z’ | -449 | 2 | $a_{16}$ | i’ | -197.73 | 0.08 |
| $a_3$ | y’ | -443 | 2 | $a_{17}$ | h’ | -193.23 | 0.08 |
| $a_4$ | x’ | -434 | 2 | $a_{18}$ | g’ | -182.74 | 0.03 |
| $a_5$ | w’ | -429 | 2 | $a_{19}$ | f’ | -162.61 | 0.05 |
| $a_6$ | v’ | -360.9 | 1 | $a_{20}$ | e’ | -155.72 | 0.05 |
| $a_7$ | u’ | -345.1 | 1 | $a_{21}$ | d’ | -138.66 | 0.05 |
| $a_8$ | t’ | -340.8 | 1 | $a_{22}$ | c’ | -130.46 | 0.05 |
| $a_9$ | s’ | -325.4 | 1 | $a_{23}$ | a’ | -98.22 | 0.03 |
| $a_{10}$ | r’ | -307.0 | 1 | $a_{24}$ | $n_2$ | -55.6 see $m_8$ Table 7 | 0.5 |
| $a_{11}$ | q’ | -298.2 | 1 | $a_{25}$ | $n_1$ | -55.6 see $m_8$ Table 7 | 0.5 |
| $a_{12}$ | p’ | -293.1 | 1 | $a_{26}$ | $m_2$ | -43.08 | 0.03 |
| $a_{13}$ | o’ | -289.7 | 1 | $a_{27}$ | $m_1$ | -41.24 | 0.05 |
| $a_{14}$ | n’ | -282.7 | 1 | $a_{28}$ | k | 0 | — |
| $a_{15}$ | j’ | -206.1 | 0.02 | | | | |

Frequency referenced to $a_{16}$ (f), R(127) 11-5, $^{127}I_2$: $f = 473\,612\,353\,604$ kHz [17]

$f(a_{28}, P(54)\ 8\text{-}4) - f(a_{16}, R(127)\ 11\text{-}5\{\text{^}(127)I_2\}) = -42.99\ (4)$ MHz [35], [36]

Ref. [35], [36], [37], [38], [39], [40], [41], [42], [43]

**Table 4**

$\lambda \approx 633$ nm $^{129}I_2$ P(69) 12-6

| $b_n$ | $x$ | $f(b_n) - f(a_{28})$/MHz | $u_c$/ MHz | $b_n$ | $x$ | $f(b_n) - f(a_{28})$/ MHz | $u_c$/ MHz |
|---|---|---|---|---|---|---|---|
| $b_1$ | b‴ | 99.12 | 0.05 | $b_{21}$ | q′ | 507.66 | 0.10 |
| $b_2$ | a‴ | 116.08 | 0.05 | $b_{22}$ | o′ | 532.65 | 0.10 |
| $b_3$ | z″ | 132.05 | 0.05 | $b_{23}$ | n′ | 536.59 | 0.10 |
| $b_4$ | s″ | 234.54 | 0.05 | $b_{24}$ | m′ | 545.06 | 0.05 |
| $b_5$ | r″ | 256.90 see $m_{28}$ Table 7 | 0.05 | $b_{25}$ | l′ | 560.94 | 0.05 |
| $b_6$ | q″ | 264.84 see $m_{29}$ Table 7 | 0.05 | $b_{26}$ | k′ | 566.19 | 0.05 |
| $b_7$ | p″ | 288.06 | 0.05 | $b_{27}$ | j′ | 586.27 | 0.03 |
| $b_8$ | k″ | 337.75 | 0.1 | $b_{28}$ | i′ | 601.78 | 0.03 |
| $b_9$ | i1″ | 358.8 | 0.5 | $b_{29}$ | h′ | 620.85 | 0.03 |
| $b_{10}$ | i2″ | 358.8 | 0.5 | $b_{30}$ | g′ | 632.42 | 0.03 |
| $b_{11}$ | f″ | 373.80 | 0.05 | $b_{31}$ | f′ | 644.09 | 0.03 |
| $b_{12}$ | d″ | 387.24 | 0.05 | $b_{32}$ | e′ | 655.47 | 0.03 |
| $b_{13}$ | c″ | 395.3 | 0.2 | $b_{33}$ | d′ | 666.81 | 0.10 |
| $b_{14}$ | b″ | 402.45 | 0.05 | $b_{34}$ | c′ | 692.45 | 0.10 |
| $b_{15}$ | a″ | 407 | 4 | $b_{35}$ | b′ | 697.96 | 0.10 |
| $b_{16}$ | z′ | 412.37 | 0.05 | $b_{36}$ | a′ | 705.43 | 0.10 |
| $b_{17}$ | y′ | 417 | 4 | | | | |

Frequency referenced to $y_{16}$ (f), R(127) 11-5, $^{127}I_2$: $f = 473\ 612\ 353\ 604$ kHz [17]

$f(a_{28}, P(54)\ 8\text{-}4) - f(a_{16}, R(127)\ 11\text{-}5\ \{^{127}I_2\}) = -42.99\ (4)$ MHz [35], [36]

Ref. [38], [41], [42], [43]

**Table 5**

$\lambda \approx 633$ nm $^{129}I_2$ R(60) 8-4

| $d_n$ | $x$ | $f(d_n) - f(a_{28})$/ MHz | $u_c$/ MHz | $d_n$ | $x$ | $f(d_n) - f(a_{28})$/ MHz | $u_c$/ MHz |
|---|---|---|---|---|---|---|---|
| $d_{23}$ | A′ | -555 | 5 | $d_{26}$ | M | -499 | 2 |
| $d_{24}$ | N | -511 | 2 | $d_{27}$ | M | -499 | 2 |
| $d_{25}$ | N | -511 | 2 | $d_{28}$ | K | -456 | 2 |

Frequency $a_{16}$ (f), R(127) 11-5, $^{127}I_2$: $f = 473\,612\,353\,604$ kHz [17]
reference $f(a_{28}, P(54)\ 8\text{-}4) - f(a_{16}, R(127)\ 11\text{-}5\ \{^{127}I_2\}) = -42.99\ (4)$ MHz [35], [36]
to

Ref. [38]

**Table 6**

$\lambda \approx 633$ nm $^{129}I_2$ P(33) 6-3

| $e_n$ | $x$ | $f(e_n) - f(e_2)$/ MHz | $u_c$/ MHz | $e_n$ | $x$ | $f(e_n) - f(e_2)$/ MHz | $u_c$/ MHz |
|---|---|---|---|---|---|---|---|
| $e_1$ | A | −19.82 | 0.05 | $e_{10}$ | J | 249 | 2 |
| $e_2$ | B | 0 | — | $e_{11}$ | K | 260 | 2 |
| $e_3$ | C | 17.83 | 0.03 | $e_{12}$ | L | 269 | 3 |
| $e_4$ | D | 102.58 | 0.05 | $e_{13}$ | M | 273 | 4 |
| $e_5$ | E | 141 | 2 | $e_{14}$ | N | 287 | 4 |
| $e_6$ | F | 157 | 2 | $e_{15}$ | O | 293 | 5 |
| $e_7$ | G | 191 | 2 | $e_{16}$ | P | 295 | 5 |
| $e_8$ | H | 208 | 2 | $e_{17}$ | Q | 306 | 6 |
| $e_9$ | I | 239 | 2 | | | | |

Frequency $a_{16}$ (f), R(127) 11-5, $^{127}I_2$: $f = 473\,612\,353\,604$ kHz [17]
reference $f(e_2, P(33)\ 6\text{-}3) - f(a_{16}, R(127)\ 11\text{-}5\ \{^{127}I_2\}) = 849.4\ (2)$ MHz [45], [46]
to

Ref. [38], [43], [45], [47]

**Table 7**

$\lambda \approx 633$ nm $^{127}I^{129}I$ P(33) 6-3

| $m_n$ | $x$ | $f(m_n) - f(a_{28})$/MHz | $u_c$/ MHz | $m_n$ | $x$ | $f(m_n) - f(a_{28})$/ MHz | $u_c$/ MHz |
|---|---|---|---|---|---|---|---|
| $m_1$ | m′ | -254 | 3 | $m_{26}$ | u″ | 212.80 | 0.05 |
| $m_2$ | l′ | -233.71 | 0.10 | $m_{27}$ | t″ | 219.43 | 0.05 |
| $m_3$ | k′ | -226.14 | 0.10 | $m_{28}$ | r″ | 256.90, see $b_5$ Table 4 | 0.10 |
| $m_4$ | j′ | -207 | 2 | $m_{29}$ | q″ | 264.84, see $b_6$ Table 4 | 0.05 |
| $m_5$ | b′ | -117.79 | 0.10 | $m_{30}$ | o″ | 299.22 | 0.05 |
| $m_6$ | p | -87.83 | 0.15 | $m_{31}$ | n″ | 312.43 | 0.05 |
| $m_7$ | o | -78.2 | 0.5 | $m_{32}$ | m″ | 324.52 | 0.03 |
| $m_8$ | n | -56, see $a_{24}$ and $a_{25}$ Table 3 | 1 | $m_{33}$ | l'' | 333.14 | 0.03 |
| $m_9$ | l | −17.55 | 0.05 | $m_{34}$ | $k_2$″ | 337.7 | 0.5 |
| $m_{10}$ | j | 12.04 | 0.03 | $m_{35}$ | $k_1$″ | 337.7 | 0.5 |
| $m_{11}$ | i | 15.60 | 0.03 | $m_{36}$ | j″ | 345.05 | 0.05 |
| $m_{12}$ | h | 33.16 | 0.03 | $m_{37}$ | h″ | 362.18 | 0.10 |
| $m_{13}$ | $g_2$ | 39.9 | 0.2 | $m_{38}$ | g″ | 369.78 | 0.03 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $m_{14}$ | $g_1$ | 41.3 | 0.2 | $m_{39}$ | e″ | 380.37 | 0.03 |
| $m_{15}$ | f | 50.72 | 0.03 | $m_{40}$ | d″ | 385 | 4 |
| $m_{16}$ | e | 54.06 | 0.10 | $m_{41}$ | x′ | 431 | 4 |
| $m_{17}$ | d | 69.33 | 0.03 | $m_{42}$ | w′ | 445 | 4 |
| $m_{18}$ | c | 75.06 | 0.03 | $m_{43}$ | v′ | 456.7 | 0.5 |
| $m_{19}$ | b | 80.00 | 0.03 | $m_{44}$ | u′ | 477.17 | 0.05 |
| $m_{20}$ | a | 95.00 | 0.03 | $m_{45}$ | t′ | 486.43 | 0.05 |
| $m_{21}$ | y″ | 160.74 | 0.03 | $m_{46}$ | s′ | 495.16 | 0.05 |
| $m_{22}$ | x″ | 199.52 | 0.03 | $m_{47}$ | r′ | 503.55 | 0.05 |
| $m_{23}$ | w″ | 205.06 | 0.05 | $m_{48}$ | p′ | 515.11 | 0.05 |
| $m_{24}$ | $v_2$″ | 207.9 | 0.5 | | | | |
| $m_{25}$ | $v_1$″ | 207.9 | 0.5 | | | | |

Frequency $a_{16}$ (f), R(127) 11-5, $^{127}I_2$: $f = 473\ 612\ 353\ 604$ kHz [17]
referenced to $f(a_{28}, \text{P(54) 8-4}) - f(a_{16}, \text{R(127) 11-5}\ \{^{127}I_2\}) = -42.99\ (4)$ MHz [35], [36]

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Ref. [38], [44], [41], [42], [43]

**Absorbing molecule $^{127}I_2$, $a_{16}$ or f component, R(127) 11-5 transition**[(1)]

[(1)] All transitions in $I_2$ refer to the $B^3\Pi\, 0_u^+ - X^1\, \Sigma_g^+$ system.

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